

Web Programming

Project Proposal

Group Members

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Title:

ShopSmart: Web Store Platform for Small Businesses

Scope:

This project aims to provide a ready-to-use e-commerce platform where small businesses can create their personalized online stores with product listings, branding options, and inventory management. Customers can browse products, add them to a cart, and place orders while tracking status. The system will include admin dashboards, analytics, and inventory alerts to help business owners manage their stores efficiently, all with a responsive, mobile-friendly UI.

● **Frontend Features**

- ◆ Home page, product listing, product detail page
- ◆ Cart system (add/remove items)
- ◆ Checkout and order confirmation
- ◆ Responsive UI for desktop & mobile

● **Backend Features**

- ◆ Admin panel for managing products, orders, inventory
- ◆ Order tracking for customers
- ◆ User accounts (optional, for order history)
- ◆ Database to store products, orders, customers

● **Additional Features**

- ◆ Inventory management (stock updates when purchase happens)
- ◆ Basic analytics (total orders, top-selling products)

● **Objectives**

- ◆ Provide small businesses a ready-to-use e-commerce platform
- ◆ Enable customers to browse, buy, and track orders online

- ◆ Help business owners manage inventory and orders efficiently

Work Break Structure:

Phase 1: Planning & Requirement Analysis

- 1.1 Identify business requirements (product types, categories)
- 1.2 Prepare site architecture & sitemap
- 1.3 Create wireframes / UI mockups

Phase 2: Database & Backend Development

- 2.1 Design database schema (products, orders, customers, inventory)
- 2.2 Implement APIs for:

- ◆ Product CRUD (Create, Read, Update, Delete)
- ◆ Order creation & tracking
- ◆ Inventory updates
- ◆ Admin authentication

- 2.3 Admin panel development (web interface for backend operations)

Phase 3: Frontend Development

- 3.1 Home page & product listing page
- 3.2 Product detail page
- 3.3 Cart & checkout page
- 3.4 Order tracking page
- 3.5 Integrate frontend with backend APIs

Phase 4: Inventory & Order Management

- 4.1 Update stock levels after purchase
- 4.2 Admin view for current inventory & alerts for low stock
- 4.3 Order status updates (pending, shipped, delivered)

Phase 5: Testing

- 5.1 Unit testing for backend API endpoints
- 5.2 UI/UX testing for frontend responsiveness
- 5.3 Functional testing for cart, checkout, admin panel
- 5.4 Integration testing

Phase 6: Deployment

- 6.1 Web server setup & database deployment
- 6.2 Optional domain configuration
- 6.3 Launch store for live usage

Phase 7: Documentation

- 7.1 User manual for admin
- 7.2 Deployment instructions
- 7.3 Demo walkthrough for ease of usage